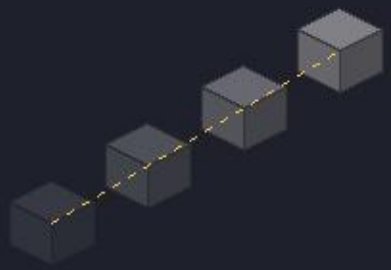
12 2D TRANSFORMATIONS

3X3 HOMOGENEOUS MATRICES APPLIED TO GAME COMPONENTS

TRANSLATION



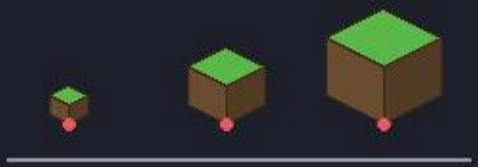
T(50,-30) X3

ROTATE ABOUT PIVOT



R(30) ABOUT SUN X12

SCALE ABOUT A POINT



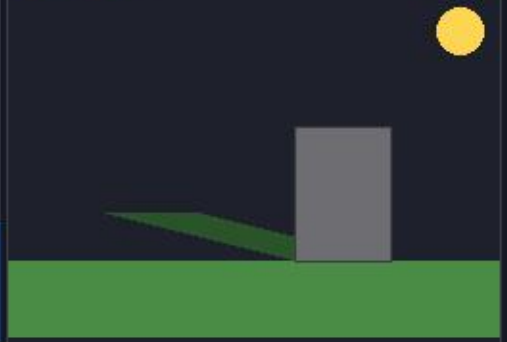
S(0.5) S(1) S(1.5)

REFLECTION



MIRROR IN THE WATER

SHEAR



SHX=1.4, SY=0.35

COMPOSITE ORDER

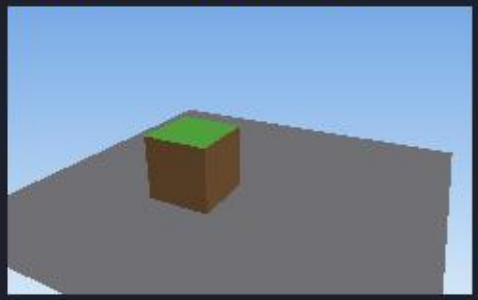



```
 0.71  0.71 110.00
-0.71  0.71   0.00
 0.00  0.00   1.00
```

13 3D TRANSFORMATIONS

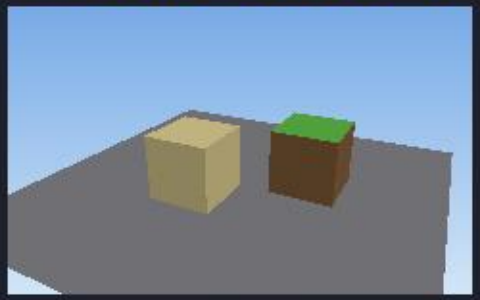
4X4 MATRICES: MODEL, VIEW (LOOK-AT), PERSPECTIVE

MODEL = IDENTITY



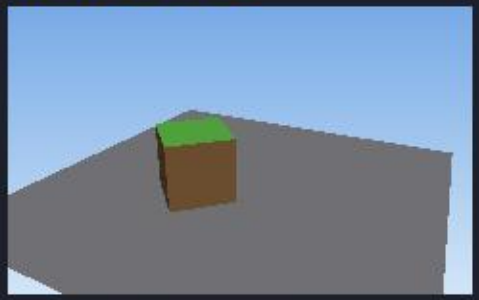
```
1.00  0.00  0.00  0.00
0.00  1.00  0.00  0.00
0.00  0.00  1.00  0.00
0.00  0.00  0.00  1.00
```

TRANSLATE (1.5,0,-1)



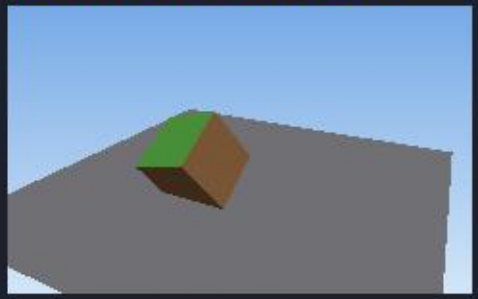
```
1.00  0.00  0.00  1.50
0.00  1.00  0.00  0.00
0.00  0.00  1.00 -1.00
0.00  0.00  0.00  1.00
```

ROTATE Y 45



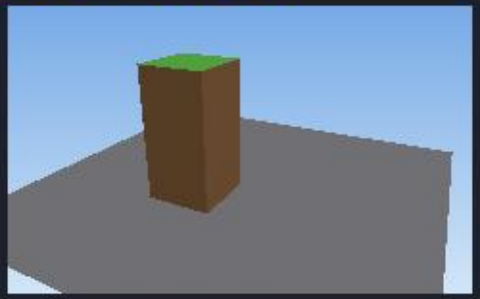
```
 0.71  0.00  0.71 -0.21
 0.00  1.00  0.00  0.00
-0.71  0.00  0.71  0.50
 0.00  0.00  0.00  1.00
```

ROTATE X 30, Z 20



```
 0.94 -0.34  0.00  0.20
 0.30  0.81 -0.50  0.20
 0.17  0.47  0.87 -0.25
 0.00  0.00  0.00  1.00
```

SCALE (1,2,1)



```
1.00  0.00  0.00  0.00
0.00  2.00  0.00  0.00
0.00  0.00  1.00  0.00
0.00  0.00  0.00  1.00
```

VIEW + PERSPECTIVE



EYE (26,17,27)

1254 VISIBLE FACES

14 ANIMATION: DAY AND NIGHT

KEYFRAMED SKY COLOUR, SUN ROTATING ABOUT A PIVOT, CLOUDS



FRAME 0



FRAME 12



FRAME 24



FRAME 36



FRAME 48



FRAME 60



FRAME 72

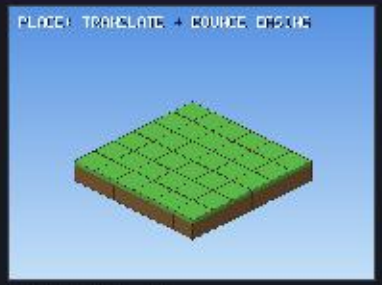


FRAME 84

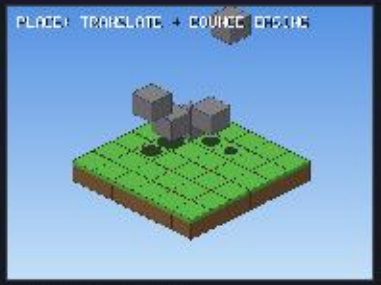
96 FRAMES - SEE OUTPUTS/CP2/GIF/DAY NIGHT.GIF

15 ANIMATION: PLACE AND REMOVE

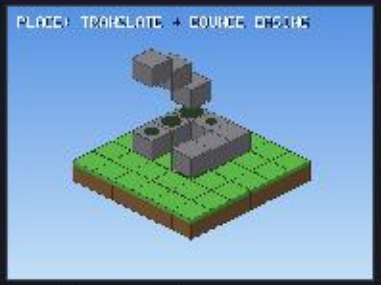
TRANSLATION WITH BOUNCE EASING, SCALE + ROTATE TO REMOVE



FRAME 0



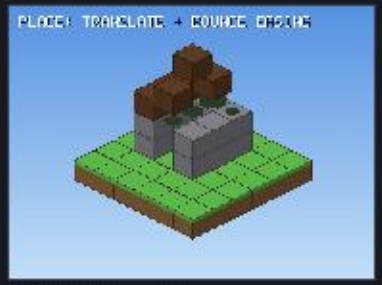
FRAME 10



FRAME 21



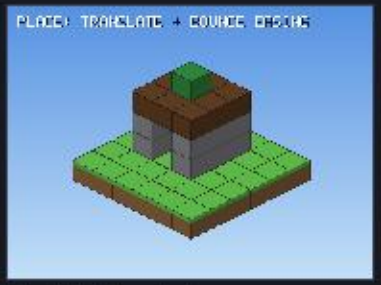
FRAME 31



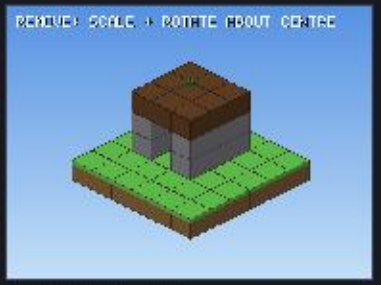
FRAME 42



FRAME 52



FRAME 63



FRAME 73

84 FRAMES - SEE OUTPUTS/CP2/GIF/BUILD.GIF

16 ANIMATION: CAMERA FLY-THROUGH

LOOK-AT CAMERA ON A SPLINE, PERSPECTIVE, Z-BUFFER, FOG



FRAME 0



FRAME 12



FRAME 24



FRAME 36



FRAME 48



FRAME 60



FRAME 72



FRAME 84

96 FRAMES - SEE OUTPUTS/CP2/GIF/FLYTHROUGH.GIF

17 ANIMATION: WALK CYCLE

ROTATION ABOUT JOINTS, BODY BOB, PARALLAX SCROLLING



FRAME 0



FRAME 8



FRAME 16



FRAME 24



FRAME 32



FRAME 40



FRAME 48



FRAME 56

64 FRAMES - SEE OUTPUTS/CP2/GIF/WALK.GIF